# Generative Modelling

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November 13, 2026

# Outline


Graphical models
- Random variables and probabilities
- Graphical model
- Time-series
- Conditional independence

Example Machine Learning Tasks
- Classification: Discriminative modelling
- Classification: Generative modelling
- Density estimation
- Discrete Bayesian Networks
- Markov model

Algorithms for latent variable models
- Gradient algorithms
- Expectation maximisation
- Bayesian Inference

Coding Exercises
- Density estimation
- Classification

Summary
- Summary

# Notational problems

In probability theory, we are dealing with functions $P$ on sets, called measures. In statistics, we are dealing with random variables. While these two can be related, it is common to use shorthand notation in statistics. If in doubt, use the following conversion table.

## Probability theory

1. Sets $A \subset \Omega$ are events.
2. Probability measure $P$ on $\Omega$.
3. $P(A) \in [0, 1]$ with $A \subset \Omega$
4. $P(A \cap C)$ the probability $\omega \in A \cap C$
5. $P(A|B)$ the probability of $A$ if $B$ is true.
6. Marginal distribution $P(A) = \sum_{i=1}^{n} \mathbb{P}(A \cap H_i)$
7. Bayes's theorem: $P(A \mid B) = P(A \cap B)/P(B)$.

## Statistics

1. Random variables $x : \Omega \to X$.
2. The measure $P$ is implicit.
3. $\mathbb{P}(x)$ for the distribution of $x$
4. $\mathbb{P}(x, y)$ the joint distribution of $x, y$.
5. $\mathbb{P}(x|y)$ the distribution of $x$ if $y$ is given.
6. Marginal distribution $\mathbb{P}(y) = \sum_{i \in X} \mathbb{P}(x = i, y)$.
7. Bayes's theorem: $\mathbb{P}(x|y) = \mathbb{P}(x, y)/\mathbb{P}(y)$

# Probability and statistics

## Probability theory

Here, we talk a lot about probability measures $P$ on some space $\Omega$. These are functions on sets, so that

- $P(\Omega) = 1$
- $P(A) \in [0, 1]$ for any $A \subset \Omega$
- $P(A \cup B) = P(A) + P(B)$ if $A \cap B = \emptyset$.

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Notes on probability.

## Measures: Functions on sets.

The idea of probability is that it is a type of measure. Measures in everyday life include length, mass, or area. The bigger the 'area' of a set, the more likely it is. It also has the same properties as areas, meaning that the total area of two separate sets is their joint area.

## Subsets of $\Omega$

When $\Omega$ is a finite set, we can consider any subset $A \subset \Omega$. However, in general we are restricted to an *algebra* of subsets. However, this is just a technical issue.

# Statistics

In statistics, we are interested in random variables $x, y, \ldots$ which are functions $x : \Omega \to X$, $y : \Omega \to Y$ etc.

- We write $\mathbb{P}(x)$ as a shorthand for *the probability distribution of* $x$.
- The distribution of $x$ arises in the following way:
  1. Sample $\omega$ from $P$
  2. Calculate $x(\omega)$.
- We can get a probability measure $P_x$ by measuring how often $x$ falls in various sets $A \subset X$:

$$P_x(A) \triangleq P(\{\omega : x(\omega) \in A\})$$

Here, $\{\omega : x(\omega) \in A\}$ is literally the set of random outcomes $\omega$ for which $x(\omega)$ is in the set $A$. To avoid writing this cumbersome expression, we simply write $\mathbb{P}(x)$ as a general shorthand. This allows us to also write

$$\mathbb{P}(x \in A) = P_x(A).$$

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Notes on random variables. The previous slide shows how we can get probability measures $P_x$ for random variables $x$. It sometimes is necessary to think about the actual measure $P_x$, but we usually do not have to assign probabilities to sets! We can just say $\mathbb{P}(x)$ loosely, without specifying whether or not we are talking about the probability of a set, or even just the related density.

# Joint distributions, multiple variables

It is still possible to define a probability measure for multiple variables. Typically, it is done as follows.

- Let $x : \omega \to X$, $y : \omega \to Y$ be two RVs.
- The measure $P_{x,y}$ has to be defined on the subsets of $X \times Y$.

$$P_{x,y}(A) = P(\{\omega : (x(\omega), y(\omega)) \in A\}$$

However, we usually just speak of the joint distribution $\mathbb{P}(x, y)$. If necessary, we may say $\mathbb{P}((x, y) \in A)$ for specific probabilities.

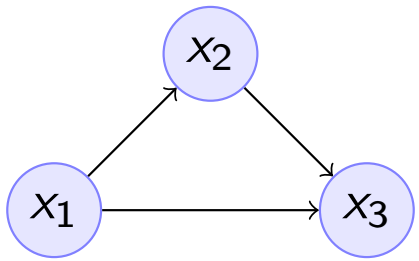
## Factoring joint distributions.

For any sequence of random variables $x_1, \ldots, x_T$, their joint distribution can be written in terms of the factors

$$\begin{aligned}\mathbb{P}(x_1, \ldots, x_T) &= \mathbb{P}(x_1)\,\mathbb{P}(x_2|x_1)\,\mathbb{P}(x_3|x_2, x_1)\cdots\mathbb{P}(x_T|x_1, \ldots, x_{T-1}) \\ &= \mathbb{P}(x_1)\prod_{t=2}^{T}\mathbb{P}(x_t|x_1, \ldots, x_{t-1})\end{aligned}$$

However, these factors can be simplified if there only few dependencies between the random variables. The structure of dependencies can be formalised in terms of graphical models.

# Graphical models




## Directed acyclic graph

A directed acyclic graph with

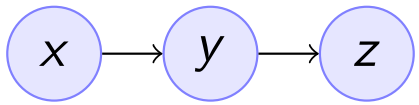
- Nodes $x_1, \ldots, x_n$
- Edges $x_i \to x_j$.

so that there are no cycles in the graph.

## Conditional independence from graphical models

- Each node of the model corresponds to a random variable
- The parent of a node are the direct dependencies of the random variable.

# Model specification




- The graphical model tells us what depends on what.
- It does not tell us how to generate data.
- We need to specify the functional relationship between variables.

## Statistics

$$x \sim f$$
$$y \mid x = a \sim g(a)$$
$$z \mid y = b \sim h(b)$$

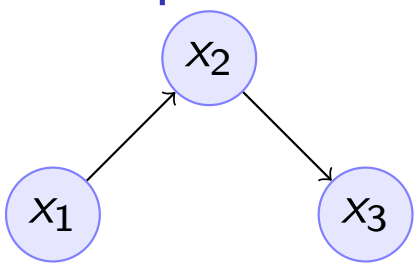
## Python

```
x = f()
y = g(x)
z = h(y)
```

# Graphical models and factorisation

- Graphical models tell us what directly depends on what
- They allow us to simplify the joint distribution of the variables
- This is formed into a product of factors, one for each variable.

## Example




This graphical model implies the factorisation

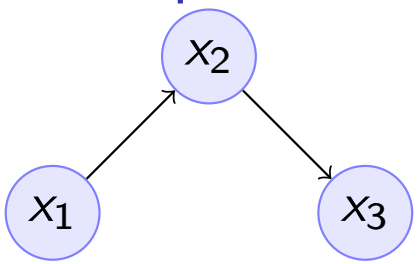
$$\mathbb{P}(x_1, x_2, x_3) = \mathbb{P}(x_3 \mid x_2, x_1)\,\mathbb{P}(x_2 \mid x_1)\,\mathbb{P}(x_1) = \mathbb{P}(x_3 \mid x_2)\,\mathbb{P}(x_2 \mid x_1)\,\mathbb{P}(x_1)$$

Notice that for each factor $\mathbb{P}(x_i \mid x_j)$, $x_j$ is the parent of $x_i$.

# Conditional independence (Chain)

- Graphical models tell us what directly depends on what
- Consequently, they also specify conditional independence

## Example




$$\mathbb{P}(x_1, x_2, x_3) = \mathbb{P}(x_3 \mid x_2)\,\mathbb{P}(x_2 \mid x_1)\,\mathbb{P}(x_1)$$

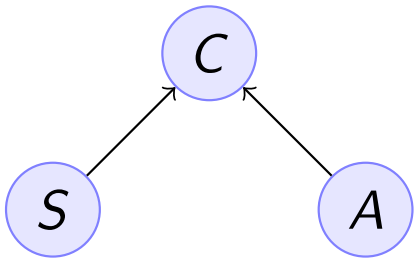
## Definition (Conditional independence)

We say that variables $x, y$ are conditionally independent given $z$ and write $x \perp\!\!\!\perp y \mid z$ if and only if

$$\mathbb{P}(x, y \mid z) = \mathbb{P}(x \mid z)\,\mathbb{P}(y \mid z)$$

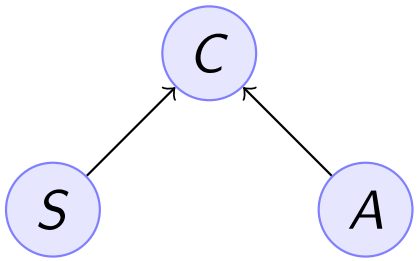
- In the above example, it holds that $x_3 \perp\!\!\!\perp x_1 \mid x_2$.

# Smoking and lung cancer (Common effect)



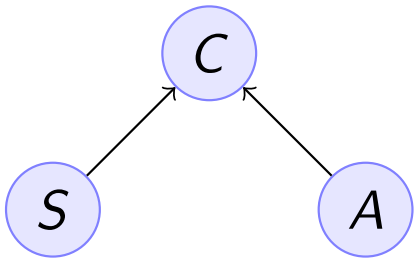

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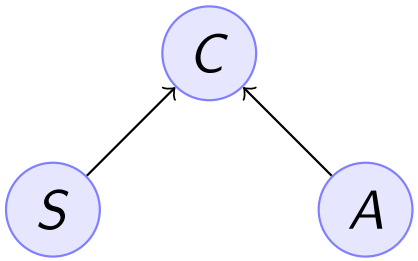



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## XOR example

- C = xor(S, A).

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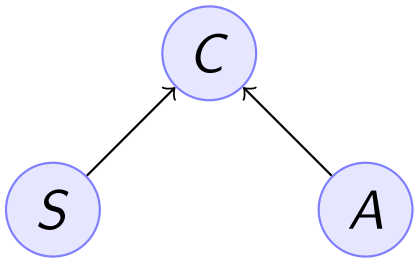



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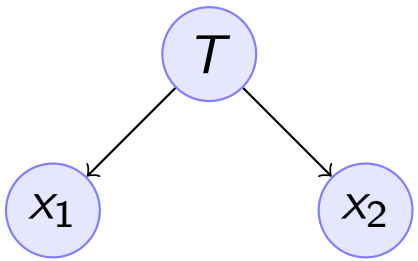



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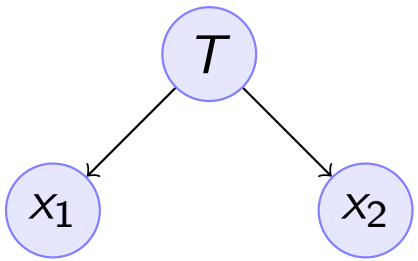
# Time of arrival at work (Common Cause)




Time of arrival at work graphical model where $T$ is a traffic jam and $x_1$ is the time John arrives at the office and $x_2$ is the time Jane arrives at the office.

- Even though $x_1, x_2$ are not independent, they become independent once you know $T$, i.e. $x_1 \perp\!\!\!\perp x_2 \mid T$.

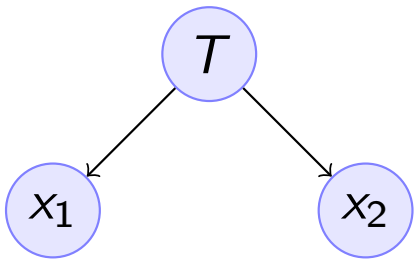
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- Proof:

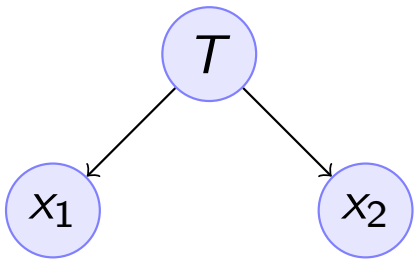
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- $\mathbb{P}(x_1, x_2, T) = \mathbb{P}(x_2 \mid T)\,\mathbb{P}(x_1 \mid T)\,\mathbb{P}(T)$ from the graph.

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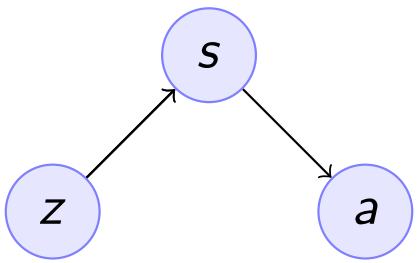
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- $\mathbb{P}(x_1, x_2, T)/\,\mathbb{P}(T) = \mathbb{P}(x_1, x_2 \mid T) = \mathbb{P}(x_2 \mid T)\,\mathbb{P}(x_1 \mid T)$. □

# School admission

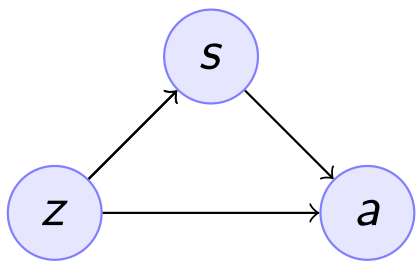
## Example

| School | Male | Female |
|---|---|---|
| A | 62 | 82 |
| B | 63 | 68 |
| C | 37 | 34 |
| D | 33 | 35 |
| E | 28 | 24 |
| F | 6 | 7 |
| Average | 50 | 27 |

- $z$: gender
- $s$: school applied to
- $a$: admission




Is admission independent of gender?




How about here?

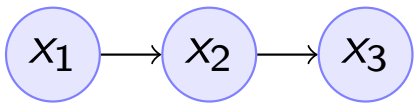
# Time-series

A sequence of data $x_1, \ldots, x_t$, is called a time series. Examples include

- Text
- Speech
- Music
- Prices
- Weather

Generally, in a time series the future observations only depend on past ones. The general idea is captured by Markov models. Even though they appear simple, they are general enough to encompass all possible models for time series.

# Markov chain



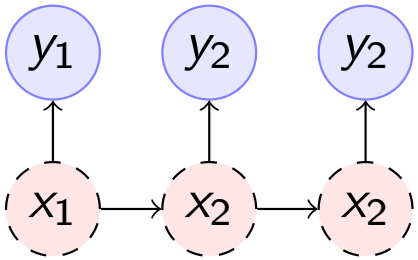

## The Markov property

In a Markov chain, the observations form a chain. In general, the following independence condition holds:

$$\mathbb{P}(x_{t+1} \mid x_t, x_{t-1}, \ldots, x_1) = \mathbb{P}(x_{t+1} \mid x_t)$$

meaning that knowing $x_t$ is sufficient to predict $x_{t+1}$, and the previous observations are not required.

# Hidden Markov model




## Hidden states

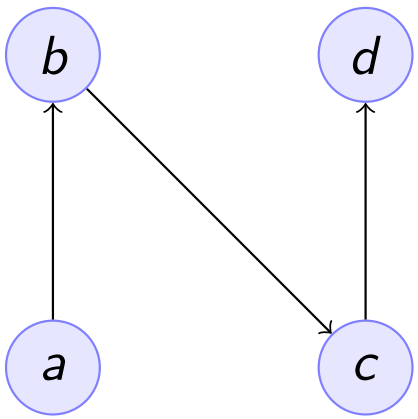
In a hidden Markov model, the observations $y_t$ are not a chain. The variable $x_t$ is not seen, and called a hidden state. The Markov property still holds for $x_t$,

$$\mathbb{P}(x_{t+1} \mid x_t, x_{t-1}, \ldots, x_1) = \mathbb{P}(x_{t+1} \mid x_t)$$

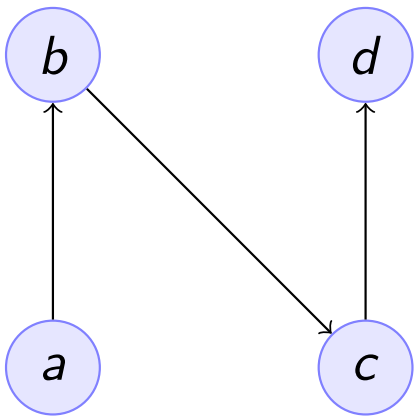
but not for $y_t$. However, if we know the current state, it is sufficient for us to predict the observations:

$$\mathbb{P}(y_t \mid y_{t-1}, \ldots, y_1, x_t, x_{t-1}, \ldots, x_1) = \mathbb{P}(y_t \mid x_t).$$

# What is the model for this graph?

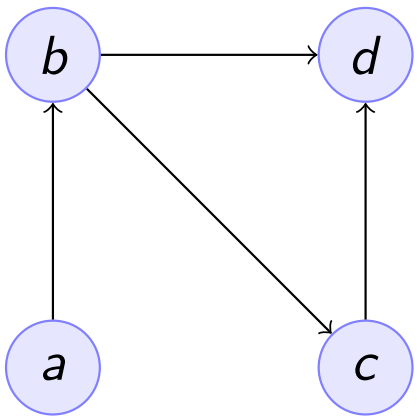


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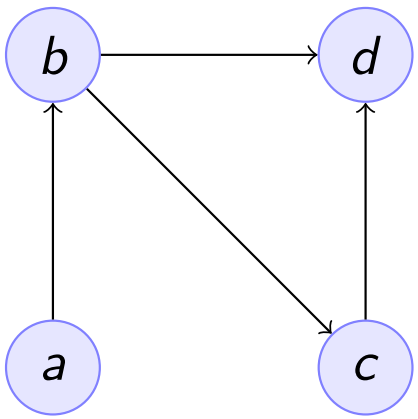



$$\mathbb{P}(a, b, c, d) = \mathbb{P}(d|c)\,\mathbb{P}(c|b)\,\mathbb{P}(b|a)\,\mathbb{P}(a)$$

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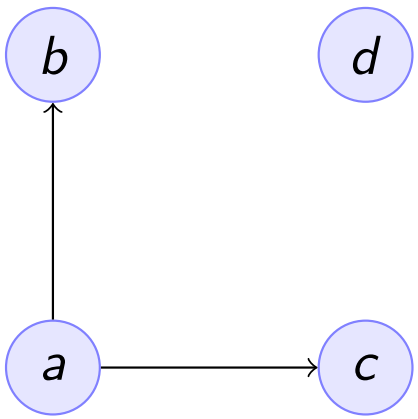


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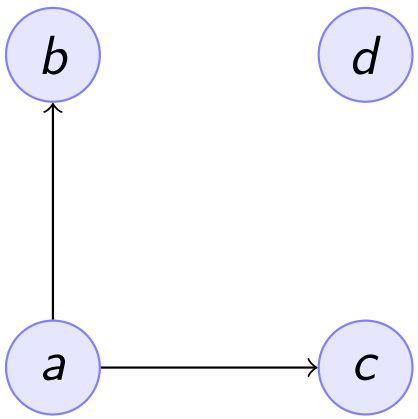



$$\mathbb{P}(a, b, c, d) = \mathbb{P}(d|b, c)\,\mathbb{P}(c|b)\,\mathbb{P}(b|a)\,\mathbb{P}(a)$$

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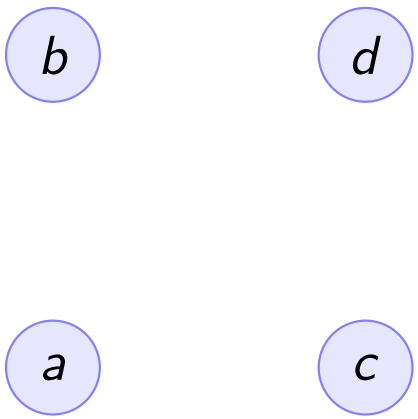


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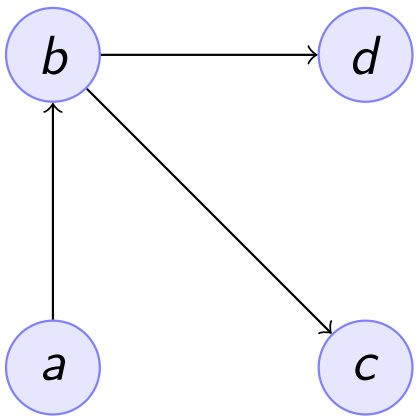

$$\mathbb{P}(a, b, c, d) = \mathbb{P}(d)\,\mathbb{P}(c|a)\,\mathbb{P}(b|a)\,\mathbb{P}(a)$$

# Draw the graph for this model



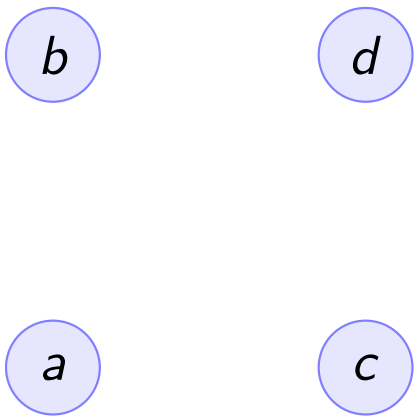

$$\mathbb{P}(a, b, c, d) = \mathbb{P}(a)\,\mathbb{P}(b|a)P(c|b)\,\mathbb{P}(d|b)$$

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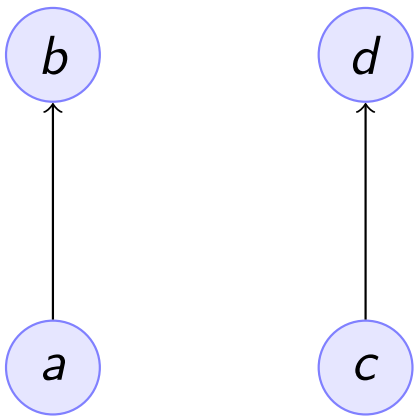

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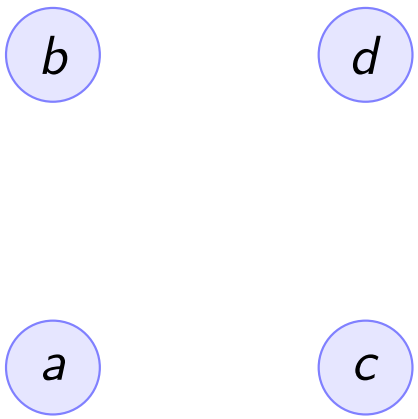

$$\mathbb{P}(a, b, c, d) = \mathbb{P}(a)\,\mathbb{P}(b|a)P(d|c)\,\mathbb{P}(c)$$

# Draw the graph for this model



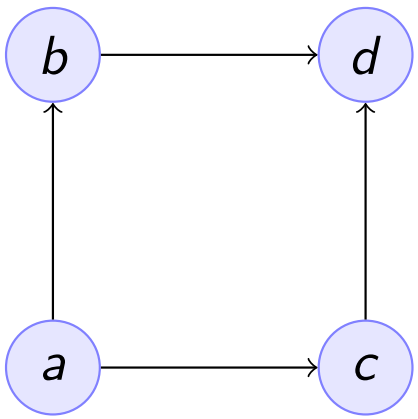

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# Draw the graph for this model




$$\mathbb{P}(a, b, c, d) = \mathbb{P}(a)\,\mathbb{P}(b|a)P(c|a)\,\mathbb{P}(d|b, c)$$

# Conditional independence (general)

- Consider variables $x_1, \ldots, x_n$.
- Let $B, D$ be subsets of $[n]$, and
- $\boldsymbol{x}_B \triangleq (x_i)_{i \in B}$ be the variables with indices in $B$.
- $\boldsymbol{x}_{-j} \triangleq (x_i)_{i \neq i}$ all the variables apart from $x_j$.

## Definition (Conditional independence)

We say $x_i$ is conditionally independent of $\boldsymbol{x}_B$ given $\boldsymbol{x}_D$ and write

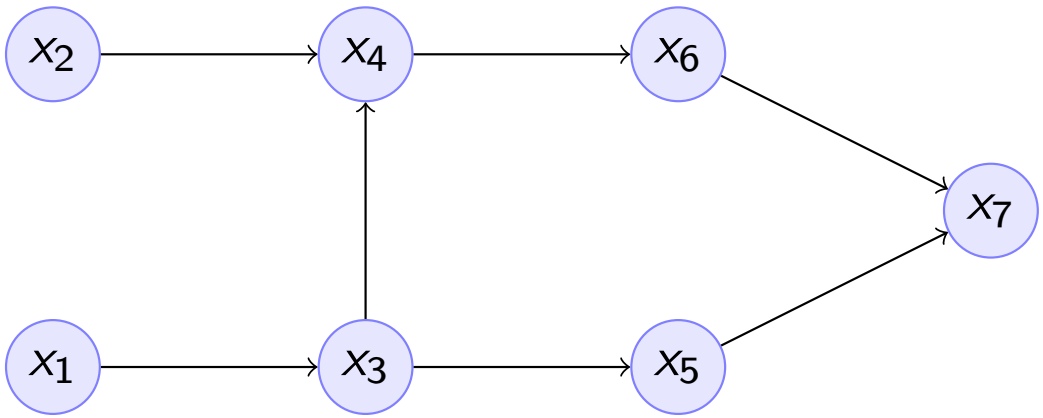
$$x_i \perp\!\!\!\perp \boldsymbol{x}_B \mid \boldsymbol{x}_D$$

if and only if:

$$\mathbb{P}(x_i, \boldsymbol{x}_B \mid \boldsymbol{x}_D) = \mathbb{P}(x_i \mid \boldsymbol{x}_D)\, \mathbb{P}(\boldsymbol{x}_B \mid \boldsymbol{x}_D).$$

- For this to hold in graphical model, $D$ must separate $i$ from $B$ in the graph.

# More complex example




In this example, we have:

$$x_7 \perp\!\!\!\perp x_1, x_2 \mid x_3, x_4$$

and

$$x_7 \perp\!\!\!\perp x_3 \mid x_4, x_5$$

Graphical models

Example Machine Learning Tasks
- Classification: Discriminative modelling
- Classification: Generative modelling
- Density estimation
- Discrete Bayesian Networks
- Markov model

Algorithms for latent variable models

Coding Exercises

Summary

# Discriminative modelling

## General idea

- Data $(x_t, y_t)$.
- Need to model $\mathbb{P}(y|x)$.
- No need to model $\mathbb{P}(x|y), \mathbb{P}(y)$.

## Examples

- Logistic regression.
- k-Nearest Neighbour
- Multi-Layer Perceptron

Discriminative modelling

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Examples
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- k-Nearest Neighbour
- Multi-Layer Perceptron

What we model. Of course, if we want to model the distribution $\mathbb{P}(x, y)$ and $x, y$ are not independent, then it should not matter whether or not we factorise $\mathbb{P}(x|y)\,\mathbb{P}(y)$ or $\mathbb{P}(x)\,\mathbb{P}(y|x)$.

However, in order for us to perform classification or regression, we only need $\mathbb{P}(y|x)$.

Then there is no need to model $\mathbb{P}(x)$ explicitly.

# Generative modelling

## General idea

- Data $(x_t, y_t)$.
- Need to model $\mathbb{P}(y|x)$.
- Model the complete data distribution: $\mathbb{P}(x|y)$, $\mathbb{P}(x)$, $\mathbb{P}(y)$.
- Calculate $\mathbb{P}(y|x) = \frac{\mathbb{P}(x|y)\,\mathbb{P}(y)}{\mathbb{P}(x)}$.

## Examples

- Naive Bayes classifier.
- Gaussian mixture model.

## Modelling the data distribution in classification

- Need to estimate the density $\mathbb{P}(x|y)$ for each class $y$.
- Need to estimate $\mathbb{P}(y)$.

Generative modelling

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Modelling the data distribution in classification
- Need to estimate the density $\mathbb{P}(x|y)$ for each class $y$.
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Generative modelling. Generative modelling generally means modelling the joint distribution of all variables. So, if we are looking at the

# The basic graphical model

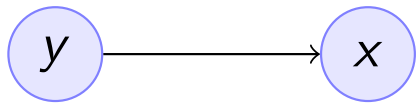
## A discriminative classification model

Here $\mathbb{P}(y|x)$ is given directly.



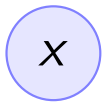

## A generative classification model

Here $\mathbb{P}(y|x) = \mathbb{P}(x|y)\,\mathbb{P}(y)/\,\mathbb{P}(x)$.

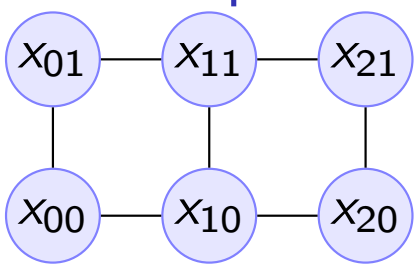



## An unsupervised generative model with no structure

Here we just have $\mathbb{P}(x)$.




## An unsupervised image model at the pixel level

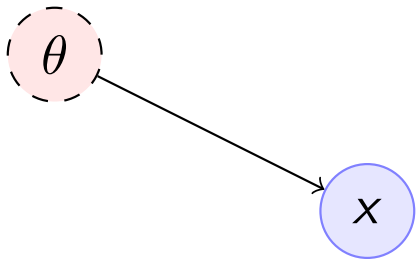



Here we model the factors $\mathbb{P}(x_{i,j}|x_{i-1,j}, x_{i,j-1}, x_{i+1,j}, x_{i,j+1})$

# Adding parameters to the graphical model

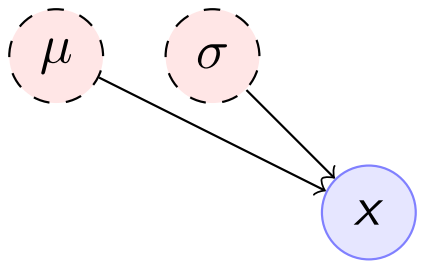
- We can also see the parameters of the distribution as (random) variables.
- We can put those random variables in the graphical model as well.
- Since the parameters are not observed, we denote them with dashed circles.
- They are a type of latent or hidden variable.

## A Bernoulli RV




Here, $x|\theta \sim \text{Bernoulli}(\theta)$

## A normally distributed variable




Here $x|\mu, \sigma \sim \text{Normal}(\mu, \sigma^2)$

# Classification: Naive Bayes Classifier

- Data $(x, y)$
- $x \in X$
- $y \in Y \subset \mathbb{N}$, $N_i$: amount of data from class $i$.

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## Separately model each class

- Assume each class data comes from a different normal distribution
- $x|y = i \sim \text{Normal}(\mu_i, \sigma_i I)$
- For each class, calculate
  - Empirical mean $\hat{\mu}_i = \sum_{t:y_t=i} x_t / N_i$
  - Empirical variance $\hat{\sigma}_i$.

# Classification: Naive Bayes Classifier

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  - Empirical variance $\hat{\sigma}_i$.

## Decision rule

Use Bayes's theorem:

$$\mathbb{P}(y|x) = \mathbb{P}(x|y)\,\mathbb{P}(y)/\,\mathbb{P}(x),$$

choosing the $y$ with largest posterior $\mathbb{P}(y|x)$.

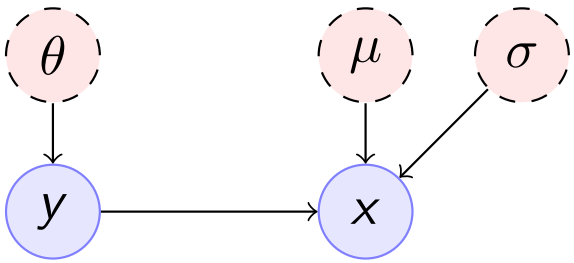
- $\mathbb{P}(x|y = i) \propto \exp(-\|\hat{\mu}_i - x\|^2/\hat{\sigma}_i^2)$

# Graphical model for the Naive Bayes Classifier

## When $x \in \mathbb{R}$

Assume $k$ classes, then

- $\mu = (\mu_1, \ldots, \mu_k)$
- $\sigma = (\sigma_1, \ldots, \sigma_k)$
- $\theta = (\theta_1, \ldots, \theta_k)$




- $y \mid \theta \sim \text{Mult}(\theta)$
- $x \mid y, \mu, \sigma \sim \text{Normal}(\mu_y, \sigma_y^2)$

# Density estimation

The simplest type of generative model is just modelling the distribution of $x$. There are a number of models for this.

## Parametric models

- Fixed histograms
- Gaussian Mixtures

## Non-parametric models

- Variable-bin histograms
- Infinite Gaussian Mixture Model
- Kernel methods

# Histograms

## Fixed histogram

- Hyper-Parameters: number of bins
- Parameters: Number of points in each bin.

## Variable histogram

- Hyper-parameters: Rule for constructing bins
- Generally $\sqrt{n}$ points in each bin.

# Gaussian Mixture Model

## Hyperparameters:

- Number of Gaussian $k$.

## Parameters:

- Multinomial distribution $\theta$ over Gaussians
- For each Gaussian $i$, center $\mu_i$, covariance matrix $\Sigma_i$.

## Algorithms:

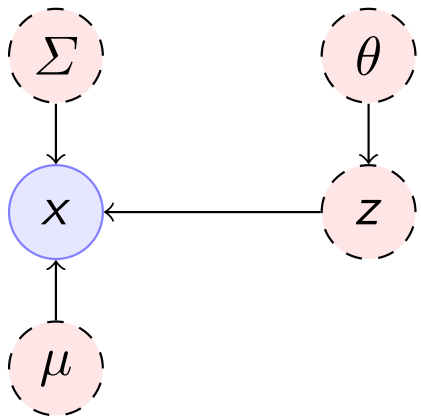
- Expectation Maximisation
- Gradient Ascent
- Variational Bayesian Inference (with appropriate prior)

# Details of Gaussian mixture models

## Model. For each point $x_t$:

- $z_t \mid \theta \sim \text{Mult}(\theta_i)$, $\theta \in \mathbb{\Delta}^k$
- $x_t | z_t = i \sim \text{Normal}(\mu_i, \Sigma_i)$.
- $\text{Mult}(\theta)$ is multinomial

$$\mathbb{P}(z_t = i \mid \theta) = \theta_i$$




- $\text{Normal}(\mu, \Sigma)$ is multivariate Gaussian

$$p(x \mid \mu, \Sigma) \propto \exp(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu))$$

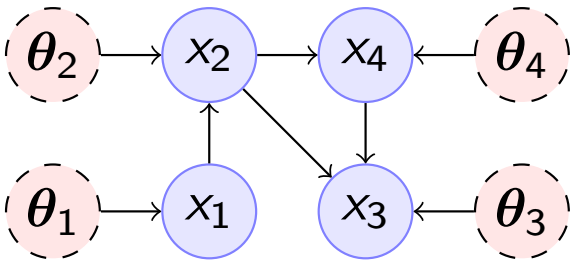
- The generating distribution is

$$p(x|\theta, \mu, \Sigma) = \sum_{z \in [k]} p(x \mid \mu_z, \Sigma_z)\, \mathbb{P}(z \mid \theta).$$

# Applications of Gaussian mixture models

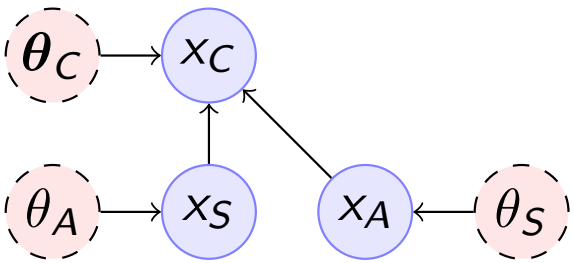
- Density estimation
- Clustering
- Used as part of a more complex model.

# Discrete Bayesian Network




- A directed acyclic graph (DAG) defined on variables $x_1, \ldots, x_n$ with each $x_n$ taking a finite number of values,
- Let $S_i$ be the indices corresponding to parent variables of $x_i$.
- $x_i \mid \boldsymbol{\theta}_i, x_{S_i} = k \sim \mathrm{Mult}(\boldsymbol{\theta}_{i,k})$.

# Example: Lung cancer, smoking and asbestos

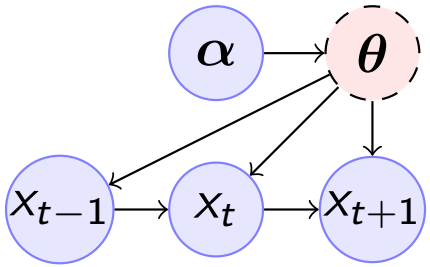



$$P_{\theta_A}(x_A = 1) = \theta_A \quad (1)$$

$$P_{\theta_S}(x_S = 1) = \theta_S \quad (2)$$

$$P_{\boldsymbol{\theta}_C}(x_C = 1 \mid X_A = j, X_S = k) = \theta_{C,j,k} \quad (3)$$

# Finite Markov models




A Markov model obeys

$$\mathbb{P}_{\boldsymbol{\theta}}(x_{k+1}|x_k,\ldots,x_1) = \mathbb{P}_{\boldsymbol{\theta}}(x_{k+1}|x_k)$$

i.e. the graphical model is a chain. We are usually interested in homogeneous models, where

$$\mathbb{P}_{\boldsymbol{\theta}}(x_{k+1} = i \mid x_k = j) = \theta_{i,j} \qquad \forall k$$

# Inference for finite Markov models

For finite Markov models, we can consider each random variable $x_{t+1}$ as having a Multinomial distribution, whose parameter depends on what the value of $x_t$. So, when $x_t = i$, then $x_{t+1}$ is multinomial with parameter $\boldsymbol{\theta}_i$.
The conjugate prior for a Multinomial parameter is the Dirichlet distribution, which is the multivariate extension of the Beta. Its density is

$$p(\boldsymbol{\theta} \mid \boldsymbol{\alpha}) \propto \prod_{i=1}^{n} \theta_i^{\alpha_i - 1}$$

## The Dirichlet-Multinomial conjugate pair

- If $x_t \in [n]$ then $x_{t+1} \mid \boldsymbol{\theta}, x_t = i \sim \mathrm{Mult}(\boldsymbol{\theta}_i)$, $\boldsymbol{\theta}_i \in \mathbb{\Delta}^n$
- Prior $\boldsymbol{\theta}_i \mid \boldsymbol{\alpha} \sim \mathrm{Dir}(\boldsymbol{\alpha})$ for all $i \in [n]$.u
- Posterior $\boldsymbol{\theta}_i \mid x_1, \ldots, x_t, \boldsymbol{\alpha} \sim \mathrm{Dir}(\boldsymbol{\alpha}^{(t)})$ with

$$\alpha_{i,j}^t = \alpha_{i,j} + \sum_{k=1}^{t} \mathbb{I}\left\{x_k = i \wedge x_{k+1} = j\right\}, \qquad \boldsymbol{\alpha}^0 = \boldsymbol{\alpha}.$$

As you can see, the parameters of the Dirichlet distribution are updated the same way as those of the beta: simply increase the counts of the observed outcomes.

Graphical models

Example Machine Learning Tasks

Algorithms for latent variable models
- Gradient algorithms
- Expectation maximisation
- Bayesian Inference

Coding Exercises

Summary

# Gradient ascent

In the following we use $\theta$ for all the parameters of the Gaussian mixture model, with $x = (x_1, \ldots, x_T)$ and $z = (z_1, \ldots, z_T)$

## Objective function

One way to estimate $\theta$ is through maximising the likelihood $L(\theta) = \mathbb{P}(x|\theta)$

## Marginalisation over latent variable

However, we need to marginalise over all values $z$

$$L(\theta) = \sum_z \mathbb{P}(z, x|\theta)$$

For $T$ data points and $k$ different values of $z_t$, there are $k^T$ vectors $z$ to sum over.

## Gradient ascent

If we can calculate the gradient of $L$, we can use gradient ascent to update our parameters:

$$\theta^{(n+1)} = \theta^{(n)} + \alpha \nabla_\theta L(\theta).$$

# Gradient calculation (method I)

Here we use the log trick: $\nabla \ln f(\theta) = \nabla f(\theta)/f(\theta)$.

$$\nabla_\theta L(\theta) = \sum_z \nabla_\theta \mathbb{P}(z, x \mid \theta) \tag{4}$$

$$= \sum_z \mathbb{P}(z, x \mid \theta) \nabla_\theta \ln \mathbb{P}(z, x \mid \theta) \tag{5}$$

$$= \sum_z \mathbb{P}(x \mid z, \theta)\, \mathbb{P}(z \mid \theta) \nabla_\theta \ln \mathbb{P}(z, x \mid \theta) \tag{6}$$

We can approximate the sum by sampling $z^{(i)}$ to obtain:[1]

$$\frac{1}{m} \sum_{i=1}^{m} \mathbb{P}(x \mid z^{(i)}, \theta) \nabla_\theta \ln \mathbb{P}(z^{(i)}, x \mid \theta), \qquad z^{(i)} \sim \mathbb{P}(z \mid \theta)$$

We can use either the exact gradient or the stochastic estimate to perform (stochastic) gradient ascent.

[1] Not really necessary for GMMs, but needed for other latent variable models.

# Gradient calculation (method II)

The problem with the previous method is that sample from the prior $\mathbb{P}(z \mid \theta)$. Let us now start with the log-likelihood and use the log trick twice

$$\nabla_\theta \ln \mathbb{P}(x \mid \theta) = \frac{\nabla \mathbb{P}(x \mid \theta)}{\mathbb{P}(x \mid \theta)} \tag{7}$$

$$= \frac{\sum_z \nabla \mathbb{P}(x, z \mid \theta)}{\mathbb{P}(x \mid \theta)} \tag{8}$$

$$= \frac{\sum_z \mathbb{P}(x, z \mid \theta) \nabla \ln \mathbb{P}(x, z \mid \theta)}{\mathbb{P}(x \mid \theta)} \tag{9}$$

$$= \sum_z \mathbb{P}(z \mid x, \theta) \nabla \ln \mathbb{P}(x, z \mid \theta). \tag{10}$$

- It does not make a practical difference whether you use method I or method II if you sum over all $z$. They are equivalent mathematically (i.e. they have the same optima) and they work exactly for the GMM.
- However, since method I optimises over the likelihood, and method II over the log-likelihood. numerical performance might be better with method II.
- Additionally, method II gives you lower-variance Monte-Carlo estimates if you sample (for more complex models), because you use $\mathbb{P}(z \mid x, \theta)$ to obtain samples from $z$, which is more informative.

# Gradient components of GMMs

Now that we looked at general latent variable models, let us see what we need to calculate for GMMs. Our parameters are $\theta = (\theta, \mu, \sigma)$.

$$\nabla_\theta \ln \mathbb{P}(z, x \mid \theta) = \sum_{t=1}^{T} \nabla_\theta \ln \mathbb{P}(z_t, x_t \mid \theta) = \sum_{t=1}^{T} \nabla_\theta \ln [\mathbb{P}(z_t \mid \theta)\, \mathbb{P}(x_t \mid z_t, \mu, \sigma)]$$

The individual terms are:

$$\nabla_\theta \ln \mathbb{P}(z_t|\theta) = \frac{1}{\mathbb{P}(z_t|\theta)} \nabla_\theta \mathbb{P}(z_t|\theta), \qquad \frac{\partial}{\partial \theta_i} = \frac{\mathbb{I}\{z_t = i\}}{\mathbb{P}(z_t|\theta)}$$

$$\frac{\partial}{\partial \mu_i} \ln \mathbb{P}(x_t \mid z_t = i, \mu, \sigma) = -\frac{\partial}{\partial \mu_i} \frac{(x_t - \mu_i)^2}{2\sigma_i^2} = \sigma_i^{-2}(x_t - \mu_i)$$

So the algorithm

- Randomly assigns $z_t$ according to the current $\theta$ estimate
- Moves the centers towards the mean indicated by the $z$.

Graphical models

Example Machine Learning Tasks

Algorithms for latent variable models
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Coding Exercises

Summary

# A lower bound on the likelihood

For any distribution $G(z)$, and specifically for $G(z) = \mathbb{P}(z|x, \theta^{(k)})$:

$$\begin{aligned}
\ln \mathbb{P}(x|\theta) &= \sum_z G(z) \ln \mathbb{P}(x|\theta) = \sum_z G(z) \ln[\mathbb{P}(x, z|\theta)/\, \mathbb{P}(z|x, \theta)] \\
&= \sum_z G(z)[\ln \mathbb{P}(x, z|\theta) - \ln \mathbb{P}(z|x, \theta)] \\
&= \sum_z G(z) \ln \mathbb{P}(x, z|\theta) - \sum_z G(z) \ln \mathbb{P}(z|x, \theta) \\
&= \sum_z \mathbb{P}(z|x, \theta^{(k)}) \ln \mathbb{P}(x, z|\theta) - \sum_z \mathbb{P}(z|x, \theta^{(k)}) \ln \mathbb{P}(z|x, \theta) \\
&\geq \sum_z \mathbb{P}(z|x, \theta^{(k)}) \ln \mathbb{P}(x, z|\theta) - \sum_z \mathbb{P}(z|x, \theta^{(k)}) \ln \mathbb{P}(z|x, \theta^{(k)}) \\
&= Q(\theta \mid \theta^{(k)}) + \mathbb{H}(z \mid x, \theta^{(k)}),
\end{aligned}$$

where

$$\mathbb{H}(z \mid x, \theta^{(k)}) = -\sum_z \mathbb{P}(z \mid x, \theta^{(k)}) \ln \mathbb{P}(z \mid x, \theta^{(k)})$$

is the entropy of $z$ for a fixed $x, \theta^{(k)}$. As this is not negative, $\ln \mathbb{P}(x|\theta) \geq Q(\theta \mid \theta^{(k)})$.

A lower bound on the likelihood

For any distribution $G(z)$, and specifically for $G(z) = \mathbb{P}(z|x, \theta^{(k)})$:

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&= \sum_z G(z) \ln \mathbb{P}(x, z|\theta) - \sum_z G(z) \ln \mathbb{P}(z|x, \theta) \\
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&\geq \sum_z \mathbb{P}(z|x, \theta^{(k)}) \ln \mathbb{P}(x, z|\theta) - \sum_z \mathbb{P}(z|x, \theta^{(k)}) \ln \mathbb{P}(z|x, \theta^{(k)}) \\
&= Q(\theta \mid \theta^{(k)}) + \mathbb{H}(z \mid x, \theta^{(k)}),
\end{aligned}$$

where

$$\mathbb{H}(z \mid x, \theta^{(k)}) = -\sum_z \mathbb{P}(z \mid x, \theta^{(k)}) \ln \mathbb{P}(z \mid x, \theta^{(k)})$$

is the entropy of $z$ for a fixed $x, \theta^{(k)}$. As this is not negative, $\ln \mathbb{P}(x|\theta) \geq Q(\theta \mid \theta^{(k)})$.

Some information theory. Information theory notation can be a bit confusing. Sometimes we talk about random variables $\omega$, and sometimes about probability measures $P$. This is context-dependent.

## Entropy

For a random variable $\omega$ under distribution $P$, we denote the entropy as

$$\mathbb{H}_P(\omega) \equiv \mathbb{H}(P) \equiv \mathbb{H}(\omega) = -\sum_{\omega \in \Omega} P(\omega) \ln P(\omega).$$

## KL Divergence

For two probabilities $P, Q$ over random outcomes in the same space $\Omega$, we define

$$D_{KL}(P \| Q) = \sum_{\omega \in \Omega} P(\omega) \ln \frac{P(\omega)}{Q(\omega)}$$

## The Gibbs Inequality

$D_{KL}(P \| Q) \geq 0$, or $\sum_x \ln P(x) P(x) \geq \sum_x \ln Q(x) P(x)$.

# From Gradient Method II to Eexpectation Maximisation

Note that Method II and EM use the same posterior distribution. The difference is that Method II performs gradient ascent on the log-likelihood, whereas EM maximises a lower bound on it.
In particular, in method II we used the gradient:

$$\nabla \ln \mathbb{P}(x \mid \theta) = \sum_z \mathbb{P}(z \mid x, \theta) \nabla \ln \mathbb{P}(x, z \mid \theta).$$

In the previous slide, we showed that

$$\ln \mathbb{P}(x \mid \theta) \geq \sum_z \mathbb{P}(z \mid x, \theta^{(k)}) \ln \mathbb{P}(x, z \mid \theta).$$

for some fixed $\theta^{(k)}$.

- The idea of the Expectation Maximisation algorithm is, instead of maximising the log-likelihood, to maximise a lower bound on it.
- In this case, this is achieved by fixing the latent distribution.
- This makes the optimisation problem simpler, and we can find an exact solution (at least for some models).

# EM Algorithm (Dempster et al, 1977)

- Initial parameter $\boldsymbol{\theta}^{(0)}$, observed data $x$
- For $k = 0, 1, \ldots$

– Expectation step:

$$Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(k)}) \triangleq \mathbb{E}_{z \sim \mathbb{P}(z|x,\boldsymbol{\theta}^{(k)})}[\ln \mathbb{P}(x, z|\boldsymbol{\theta})] = \sum_z [\ln \mathbb{P}(x, z|\boldsymbol{\theta})] \, \mathbb{P}(z \mid x, \boldsymbol{\theta}^{(k)})$$

– Maximisation step:

$$\boldsymbol{\theta}^{(k+1)} = \arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)}).$$

See *Expectation-Maximization as lower bound maximization, Minka, 1998*

# Minorise-Maximise

EM can be seen as a version of the minorise-maximise algorithm

- $f(\boldsymbol{\theta})$: Target function to maximise
- $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)})$: surrogate function

## $Q$ Minorizes $f$

This means surrogate is always a lower bound so that

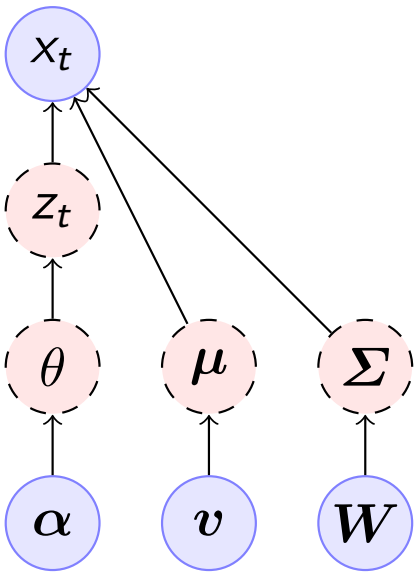
$$f(\boldsymbol{\theta}) \geq Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)}), \qquad f(\boldsymbol{\theta}^{(k)}) = Q(\boldsymbol{\theta}^{(k)}|\boldsymbol{\theta}^{(k)}),$$

## Algorithm

- Calculate: $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)})$
- Optimise: $\boldsymbol{\theta}^{(k+1)} = \arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)})$.

# Bayesian inference for the GMM

We saw, for the Beta-Bernoulli and Dirichlet-Multinomial models, that a conjugate prior turns Bayesian inference into a simple parameter update. The same idea extends to the GMM, by placing a prior on every parameter.




$$\mu_i \mid \boldsymbol{v} \sim \text{Normal}(\boldsymbol{v}, I), \tag{11}$$
$$\Sigma_i \mid \boldsymbol{W} \sim \text{Wish}(\boldsymbol{W}), \tag{12}$$
$$\theta \mid \boldsymbol{\alpha} \sim \text{Dir}(\boldsymbol{\alpha}), \tag{13}$$
$$z_t \mid \theta \sim \text{Mult}(\theta), \tag{14}$$
$$x_t \mid \mu, \Sigma, z_t = i \sim \text{Normal}(\mu_i, \Sigma_i). \tag{15}$$

## Exercise

Which of the five lines above, if any, would have a closed-form conjugate update if all the others were held fixed? Compare with the Beta-Bernoulli and Dirichlet-Multinomial cases covered earlier.

Graphical models

Example Machine Learning Tasks

Algorithms for latent variable models

Coding Exercises
- Density estimation
- Classification

Summary

# GMM versus histogram

- Generate some data $x$ from an arbitrary distribution in $\mathbb{R}$.
- Fit the data with a histogram for varying numbers of bins
- Fit a GMM with varying numbers of Gaussians
- What is the best fit? How can you measure it?

# GMM Classifier

## Base class: sklearn GaussianMixtureModel

- *fit()* only works for Density Estimaiton
- *predict()* only predicts cluster labels

## Problem

- Create a GMMClassifier class
- *fit()* should take X, y, arguments
- *predict()* should predict class labels
- Hint: Use $predict_{proba}()$ and multiple GMM models

# Summary

- CPTs store conditional probabilities for discrete random variables.
- Conditional independence reduces the number of parameters.
- Graphical models represent specific conditional independence structures.
- Models can also contain *latent* variables.
- Latent variables can be parameters, cluster IDs, etc

# Example graphical models

- Naive Bayes
- Gaussian Mixture Model
- Linear regression
- Logistic regression
- Many other neural networks (those with a probabilistic interpretation).

Next: fairness!